UNIVERSITÉ PARIS 1
PANTHÉON SORBONNE

JAPAN'S LOST MULTIPLIER

Monetary Endogeneity Across Unconventional Policy Regimes through Functional Coefficient FAVAR and Inverse Reinforcement Learning

Author: Elia Landini
Supervisor: Prof. Jean-Bernard Chatelain

MRes in Financial Economics
Academic Year 2025–2026

elia.landini@etu.univ-paris1.fr

JEL: C32, C61, E12, E51

Structure

Part I - Foundations

1 - Introduction & Research Question
2 - Exogenous Money
3 - Endogenous Money

Part II - Empirical Approach

4 - Data Retrieval & Variable Construction
5 - FC-FAVAR Model
6 - Three-Agent IRL Model

Part III - Results

7 - FC-FAVAR Results
8 - IRL Results (Global, Exo, YCC-Restricted)

Part IV - Closing

9 - Conclusions & Implications
10 - Acknowledgements

Introduction & Research Hypothesis

The Japanese Puzzle

- BoJ balance sheet: ¥158tn (2013) $\rightarrow$ ¥553tn (2018), yet M3 growth stayed at 2–3% p.a.
- Ex-post multiplier $m = M/H \rightarrow 0$: reserve injections did not generate broad money
- $\rho(\text{Total Debt}, M3)^{\text{JP}}_{1980\text{–}2019} = 0.987$ (Yamaguchi, 2022): *debt*, not base, explains M3

Research Question

Does money creation follow *exogenous* logic ($H \rightarrow M \rightarrow P$ via multiplier) or *endogenous* logic (credit demand $\rightarrow$ deposits $\rightarrow$ reserves)? Is this *regime-contingent*?

JEL: C32, C61, E12, E51

Central Hypothesis H_0

Private credit *Granger-causes* M3; the reverse does *not*:

$$\Delta M3_t = \alpha + \sum_{k=1}^{p} \beta_k \, \Delta \mathrm{Cr}_{t-k} + \varepsilon_t$$

$H_0: \beta_k \neq 0$ and $\gamma_k = 0$ in the reverse.

Evaluated at four regime centroids z_j via rolling FC-FAVAR; reproduced bottom-up via three-agent IRL.

Literature I — Exogenous Money

Quantity Theory — Fisher (1911)

$MV = PT \Rightarrow M = PT/V$

CB controls M; with V, T stable: $\Delta M \propto \Delta P$.

Money Multiplier — Brunner & Meltzer (1968)

$M = m \cdot H, \quad m = (1 + c)/(c + r)$

H: monetary base; c: cash ratio; r: reserve ratio.

BoJ controls $H \Rightarrow M$ is supply-determined.

Liquidity Preference — Keynes (1936)

$L(r, Y) = M/P, \quad \partial L/\partial r < 0, \ \partial L/\partial Y > 0$

Fixed supply absorbed by demand; r clears market.

ZLB: $\partial L/\partial r \to \infty$ — liquidity trap.

IS-LM & AS-AD — Hicks (1937)

$M/P = L(r, Y) \Rightarrow$ BoJ shifts LM exogenously

- $\uparrow r \Rightarrow \downarrow I \Rightarrow \downarrow Y$
- ZLB: liquidity trap severs transmission
- Long run: money neutral (AS-AD)

Japan: $\uparrow H \not\Rightarrow \uparrow M3 \not\Rightarrow \uparrow P$ — *empirically falsified.*

Literature II — Endogenous Money

Causal Inversion — Kaldor (1982)

Loans $\rightarrow$ Deposits $\rightarrow$ Reserves

CB accommodates ex-post; money is endogenous to credit.

Horizontalism — Moore (1988)

$r = r^*$ (CB-set), L^s perfectly elastic at r^*

Quantity solely determined by borrower demand.

Endogenous IS-LM — Yamaguchi (2022)

$M = f(D_h, D_p, D_g)$, $\rho^{JP}_{M,\mathrm{Debt}} = 0.987$

IS & LM compress jointly under demand contraction.

Structuralism — Palley (2013)

$r^L = r^* + \phi(\kappa, \mathrm{NIM}, \delta)$

κ: capital ratio; NIM: net interest margin; δ: default risk.

Endogeneity *degree* varies with institutional structure.

Degree of Endogeneity — Palley (2013)

$\varepsilon_{L^d} = (\partial L^d / \partial r^L) \cdot (r^L / L^d)$

At ZLB: $\varepsilon_{L^d} \rightarrow 0$, credit demand inelastic, transmission channel severed.

Data - Variable Construction & Stationarity

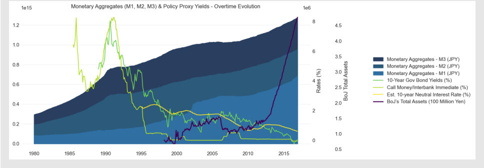
Panel: Japan, monthly, 1960M1–2026M3 (905 obs., 24 variables)
Sources: FRED, Eurostat, World Bank, Yahoo Finance, DBnomics

Category	**Variables**
Monetary agg.	M1, M2, M3 (SA, JPY)
Credit agg.	Corp., HH, Gov. (% GDP)
BoJ balance sheet	Total assets, QE/YCC proxy
Interest rates	10Y JGB, call rate, r^n_{1Y}, r^n_{10Y}, loan rate, deposit rate
Prices & activity	HICP, real GDP, VIX
Exchange rates	USD/JPY (CPI-based REER)

Stationarity: ADF + Phillips-Perron; I(1) vars. first-differenced (log-levels for price/quantity); regime indicator z_t constructed as composite PC of policy instruments.

QE 2001–2006	**QQE** 2013+
NIRP 2016+	**YCC** 2016–2024

Figure: Monetary Aggregates & Policy Yields



Econometric Approach I — FC-FAVAR Specification

Step 1 — Factor Extraction

We compress $N = 24$ transformed variables into $K = 5$ latent factors via PCA.

$$\mathbf{X}_t = \mathbf{\Lambda}\mathbf{F}_t + \mathbf{e}_t$$

The retained factors are the five principal components of the transformed informational panel.

Step 2 — FAVAR System

The restricted vector is $Y_t = (\ln M3_t,\ CPNF_t,\ CM_t,\ F1_t)'$.

$$Y_t = A(L)Y_{t-1} + B(L)F_{t-1} + u_t$$

The policy proxy enters through the identified VAR block; Cholesky ordering places credit before money and the policy rate last.

Step 3 — Functional Coefficient Extension

Coefficients vary smoothly with the regime indicator z_t.

$$\mathbf{\Phi}(L, z_t) = \sum_{j=1}^{p} \mathbf{\Phi}_j(z_t)L^j$$

Local estimation uses weighted least squares with kernel weights around each evaluation point z_0.

$$\mathbf{\Phi}(z_0) = \arg\min_{\mathbf{\Phi}} \sum_{t=1}^{T} \|\mathbf{y}_t - \mathbf{\Phi}(z_t)\mathbf{x}_t\|^2 K_h(z_t - z_0)$$

Bandwidth h selected by leave-one-out cross-validation, following Cai and Liu (2021).

Econometric Approach II — Granger Tests & Rolling IRF

Regime-Specific Granger Causality

At each regime centroid $z_0 \in \{z_1, z_2, z_3, z_4\}$, extract the local coefficient matrices and test whether lagged credit terms jointly predict broad money.

Wald null: $H_0 : \beta_j^{\mathrm{Cr}\to M3}(z_0) = 0, \ \forall j = 1, \ldots, p$.

z_1	Conventional / bubble (pre-2001)
z_2	QE (2001–2006)
z_3	QQE / NIRP (2013+)
z_4	YCC (2016–2024)

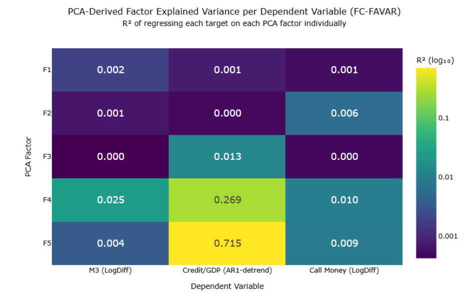
Rolling IRF Surface

Five-year rolling windows; regime-indexed responses are computed by evaluating the companion matrix at each local point z_0.

$$\mathrm{IRF}_h(z_0) = \mathbf{C}(z_0)^h \mathbf{P}(z_0), \qquad h = 0, 1, \ldots, H$$

This yields a continuous surface of credit-shock responses across horizons and policy states.

Figure: PCA-Derived Factor Explained Variance



IRL Approach I — State Variables & Transition Dynamics

State Vector & Actions

The macroeconomic state is a nine-dimensional vector collecting output, inflation, the policy rate, capital stress, default risk, bank credit, NIM, household credit demand, and household debt proxy.

$$\mathbf{s}_t = \left(y_t,\ \pi_t,\ r_t,\ \mathrm{CapStress}_t,\ \delta_t,\ L_t^b,\ \mathrm{NIM}_t,\ L_t^{hh},\ D_t^{hh}\right)$$

Credit efficiency is the binding constraint between bank credit supply and household demand; recessionary pressure is activated only for negative output gaps.

OLS Transition System

State dynamics are governed by five forward OLS regressions estimated on Japanese monthly data.

$$\begin{aligned}
y_{t+1} &= \alpha_y + \rho_y y_t + \beta_{CE}\mathrm{CE}_t + \beta_{CS}\mathrm{CapStress}_t + \varepsilon_t^y \\
\pi_{t+1} &= \alpha_\pi + \rho_\pi \pi_t + \lambda_y y_t + \varepsilon_t^\pi \\
\delta_{t+1} &= \alpha_\delta + \rho_\delta \delta_t + \lambda^- y_t^- + \lambda_b L_t^b + \varepsilon_t^\delta
\end{aligned}$$

$$\mathrm{CapStress}_{t+1} = \alpha_C + \rho_C \mathrm{CapStress}_t + \gamma_\delta \delta_t - \gamma_n \mathrm{NIM}_t + \varepsilon_t^C$$

$$D_{t+1}^{hh} = \alpha_h + \rho_h D_t^{hh} + \eta_L L_t^{hh} - \eta_r r_t + \varepsilon_t^h$$

IRL Approach II — Inferred ω & Reward Functions

IRL Problem (Ng & Russell, 2000)

Given observed trajectories $\mathcal{D} = \{\mathbf{s}_t, a_t\}_{t=1}^{T}$, we recover the latent linear reward:

$$R(\mathbf{s}, a; \omega) = \omega^\top \phi(\mathbf{s}, a)$$

BoJ Reward

$$R^{\mathrm{BoJ}} = -\omega^\pi(\pi_t - \bar{\pi})^2 - \omega^y y_t^2 - \omega^\kappa \kappa_t^-$$

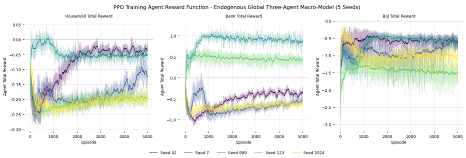
Bank Reward

$$R^{\mathrm{bank}} = \omega^{\mathrm{NIM}}\mathrm{NIM}_t - \omega^\delta \delta_t - \omega^\kappa \kappa_t^-$$

Household Reward

$$R^{hh} = \omega^C C_t - \omega^L \left(r_t^L L_t^{hh}\right)$$

Figure: PPO Training Agent Reward Function (5-seed)



Action Controls

- **BoJ:** $a_t^{\mathrm{BoJ}} = (\Delta r_t, \Delta QE_t)$
- **Bank:** $a_t^{\mathrm{bank}} = \Delta L_t^b$
- **Household:** $a_t^{hh} = \Delta L_t^{hh}$

IRL Approach III — Counterfactual & YCC-Restricted Models

Exogenous Counterfactual

We replace the bank's strategic lending rule with a mechanical money-multiplier specification:

$$L_t^b = m \cdot \left(QE_t - \underline{QE}\right), \qquad m = 4.0$$

The bank actor is switched off, receives no reward signal, and no longer updates its policy. Credit supply becomes a pure function of BoJ reserve injection, independent of household demand.

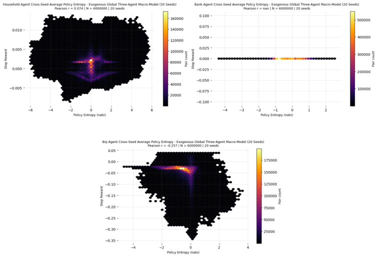
YCC-Restricted Endogenous Model

Here, we re-estimate the full environment only on the YCC subsample and re-infer the regime-specific omega vectors from that period alone:

$$\Delta r_t \geq -0.01, \qquad \Delta QE_t \geq 0$$

The YCC-restricted model is estimated on 2016-02 to 2022-01 and tests whether the YCC incentive structure removes the coordination failure found in the global endogenous model.

Figure: PPO Training Agent Reward Function (5-seed)



PPO Training Architecture

Gaussian actor-critic PPO with two hidden layers of 64 units, 5,000 episodes, 60-step horizon, and 20 random seeds for robustness. Cross-seed trajectories are reported as 50-window moving averages.

FC-FAVAR Results I — Granger Causality Across Regimes

Regime-Dependent Granger Evidence

Regime	**Credit → M3**	**M3 → Credit**
z_1: Conventional / bubble (pre-2001)	no significant effect	no significant effect
z_2: QE (2001-2006)	$F = 0.285$, not significant	no significant effect
z_3: QQE / NIRP (2013+)	**strongly significant**	negative signal, not significant
z_4: YCC (2016-2023)	**strongly significant**	negative signal, not significant

Interpretation

- Regimes 1-2 show informational inertia: neither direction is active, consistent with post-bubble balance-sheet repair and weak credit transmission.
- Regimes 3-4 activate a clear unidirectional Credit→M3 channel, which matches exactly the causal inversion codified by Kaldor (1982) and the endogenous-money tradition.
- The reverse M3→Credit direction does not become significant; its negative sign under QQE and YCC is consistent with BoJ asset purchases raising M3 while compressing bank NIM and weakening lending incentives.

FC-FAVAR Results II — Rolling IRF Surfaces

Figure: Credit → M3 Surface

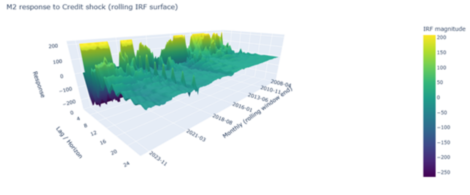
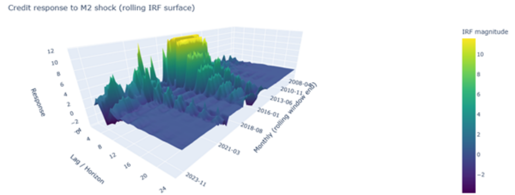


Figure: M3 → Credit Surface



Credit→M3 is near zero before QQE, then turns sharply positive at short horizons after 2013, consistent with the contemporaneous deposit-creation logic of Moore 1988. M3→Credit stays essentially flat except during the GFC window, where both variables compress jointly rather than one causing the other, which is consistent with endogenous money. We note that the YCC segment shows a more persistent medium-horizon positive response, suggesting that demand-led expansion is more durable than the earlier QQE phase.

IRL Results I — Endogenous Global Model

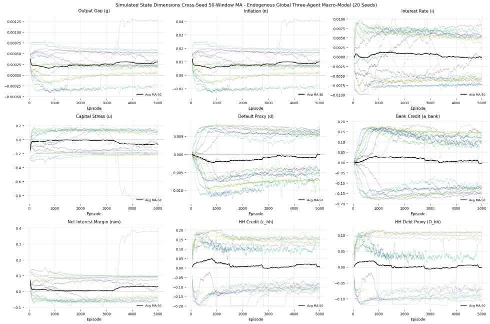
Bimodal Attractor Structure

- **Low-activity attractor:** the bank learns aggressive credit rationing, household demand weakens, and the system remains near-zero credit with a negative output gap and below-zero inflation.
- **Activated attractor:** credit demand remains moderately positive, NIM stabilises, and inflation partially recovers toward zero.
- The low-activity equilibrium is visited more often, consistent with the informational inertia detected by the FC-FAVAR in Regimes 1–2.

Coordination Failure

Bank rationing and weak household demand are mutually self-reinforcing; elastic reserve supply by the BoJ is not sufficient to move the system unilaterally toward the activated equilibrium.

Global Endogenous Model Trajectories



IRL Results II — Exogenous Counterfactual & YCC-Restricted

Exogenous Counterfactual

The endogenous bank decision is replaced by a passive money-multiplier rule, so the bank receives no reward signal and its entropy collapses to zero.

- Bank credit converges to a mechanically positive level, but only because the multiplier forces supply independently of household demand
- The default proxy accumulates silently once the rationing brake is removed
- NIM compression under the low-rate environment remains unmitigated, feeding higher capital stress and larger cross-seed dispersion
- Inflation is lower and less convergent across seeds despite higher simulated credit
- The simulation reproduces the NIM–capital–lending doom loop of Japan's post-bubble banking crisis

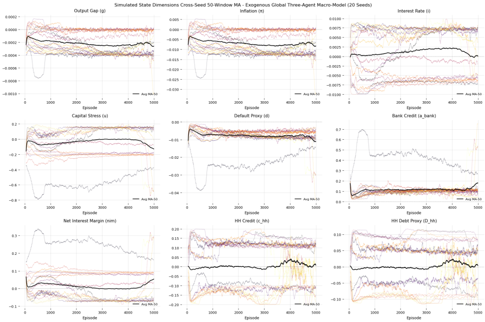
YCC-Restricted Endogenous Model

The environment is re-estimated only on the YCC subsample, and the omega vectors are re-inferred on 2016-02 to 2022-01 data.

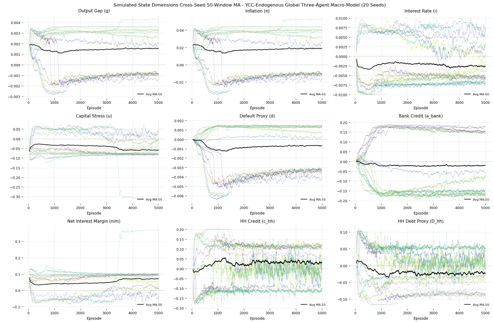
- The coordination failure becomes structurally absent as the bank's dominant conditioning variable shifts from default risk to household credit demand
- The output gap stabilises around a slightly positive average and inflation converges to a persistently positive level without sustained acceleration
- Bank and household credit converge to small positive levels with tight cross-seed dispersion from roughly episode 2,000 onward
- This closes the loop with the FC-FAVAR evidence: the activated Credit→M3 channel in the YCC regime is reproduced bottom-up through incentive recalibration

IRL Results III — Exogenous Counterfactual & YCC-Restricted Trajectories

Figure: Exogenous Counterfactual Trajectories



YCC-Restricted Endogenous Model Trajectories



Conclusions

Main Findings

1. **Regime-contingent endogeneity:** monetary endogeneity is not universal; it activates when bank capital adequacy, NIM stability, and default risk normalisation are restored.
2. **FC-FAVAR:** no significant Credit→M3 channel in Regimes 1–2; strong activation in Regimes 3–4; the reverse direction remains non-significant with a negative signal.
3. **IRL endogenous:** the global model reproduces a bimodal coordination failure in which bank rationing and weak household demand mutually reinforce a low-activity attractor.
4. **IRL exogenous:** replacing strategic lending with a mechanical multiplier generates the NIM–capital–lending doom loop observed in Japan's post-bubble banking stress.
5. **IRL YCC-restricted:** recalibrating incentives to the YCC period removes the coordination failure and reproduces the demand-led credit expansion identified in the FC-FAVAR stage.

Policy Implications

- Reserve injections alone are insufficient; credit activation depends on recapitalisation, NIM stabilisation, and default-risk normalisation.
- Yield management under YCC appears more effective than pure quantity-based base expansion for reactivating the credit-money channel.
- The findings support Palley's structuralist formulation over a pure horizontalist reading of endogenous money.

Limitations & Future Research

- Fixed regime centroids in the FC-FAVAR suppress within-regime heterogeneity.
- The toy economy abstracts from fiscal policy, public debt, and the foreign sector.
- A natural extension is a four- or five-agent IRL model with government and exchange-rate dynamics.

THANK YOU

Questions and comments are most welcome.

Repository:

`github.com/EliaLand/IRL-FAVAR_money_endogeneity`